

STATUS of the HALL A MØLLER POLARIMETER

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for the Moller Polarimeter Collaboration:

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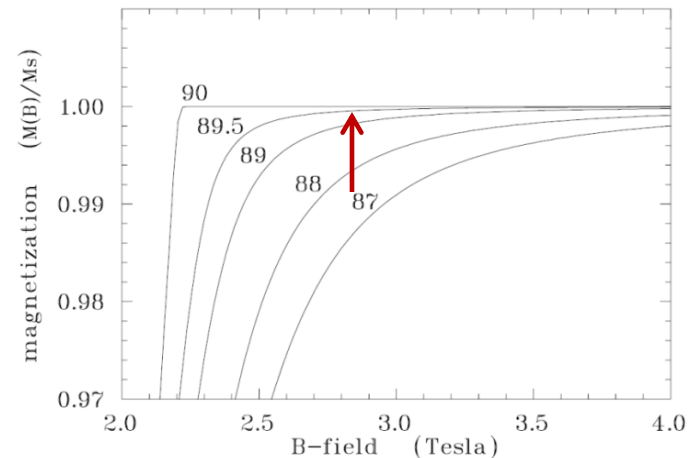
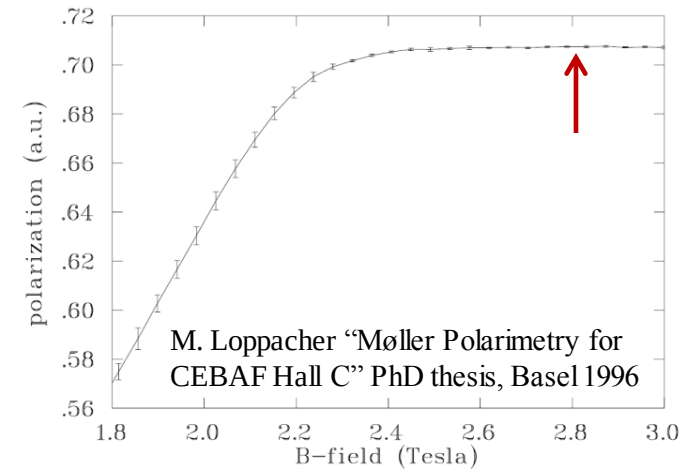
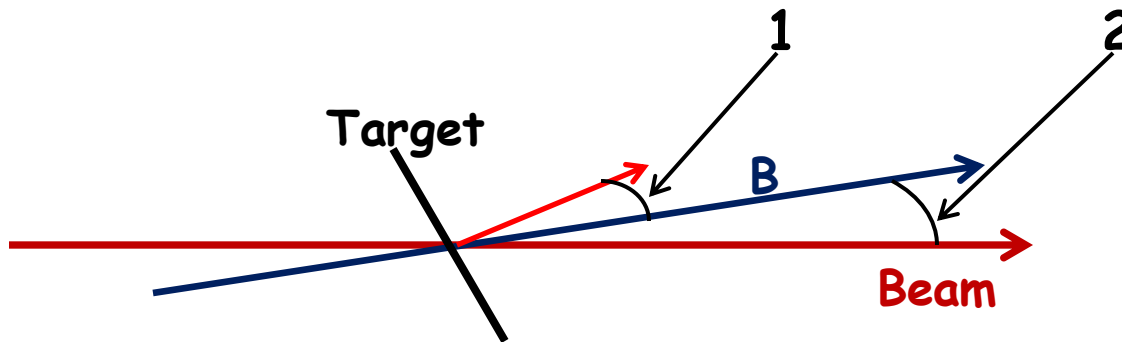
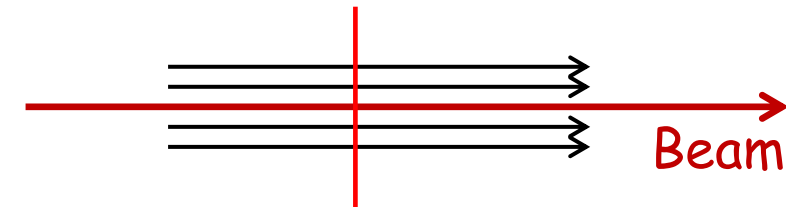
Temple University

"BRUTE FORCE" TARGET

Two alignments is a key issue of "brute force" target:

- 1) *Target vs. magnetic field* → *saturation, discrepancy between targets*
 - better target alignment or/and stronger magnetic field
- 2) *Magnetic field vs. beam direction* → *acceptance: analyzing power, Levchuk-effect, transverse polarization component*
 - test with a beam deflection on downstream BPMs,
 - better magnet alignment

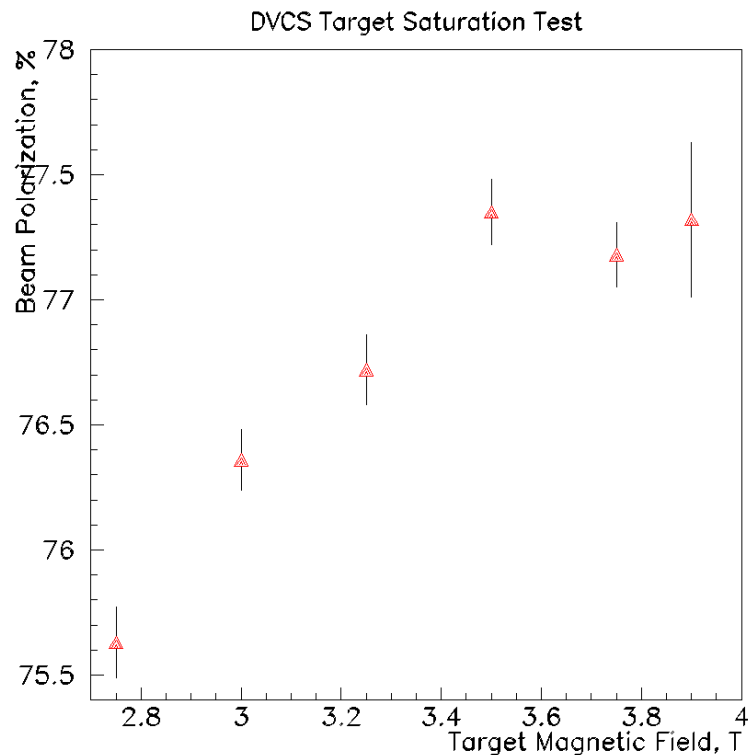
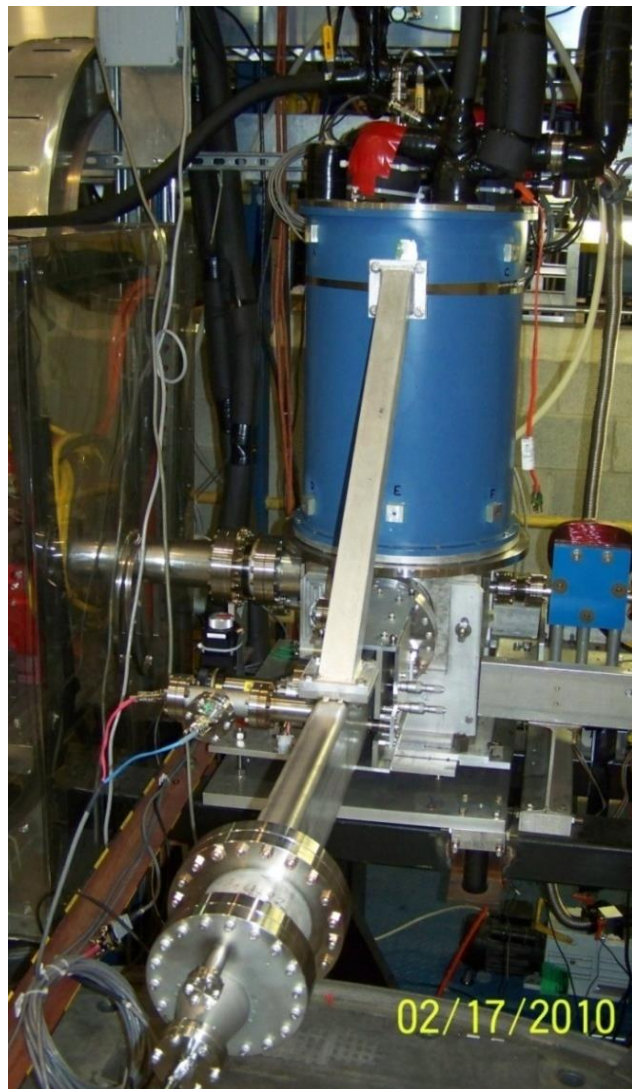
Target >2.5T



HALL A "BRUTE FORCE" TARGET 2010

Problems:

- SC magnet required liquid N and low pressure liquid He,
- Complex and expensive operation,
- Daily refill with liquid He,
- 4T max Field, >3.5T for the target saturation,
- No remote target alignment,
- No success with magnet field alignment along the beam,

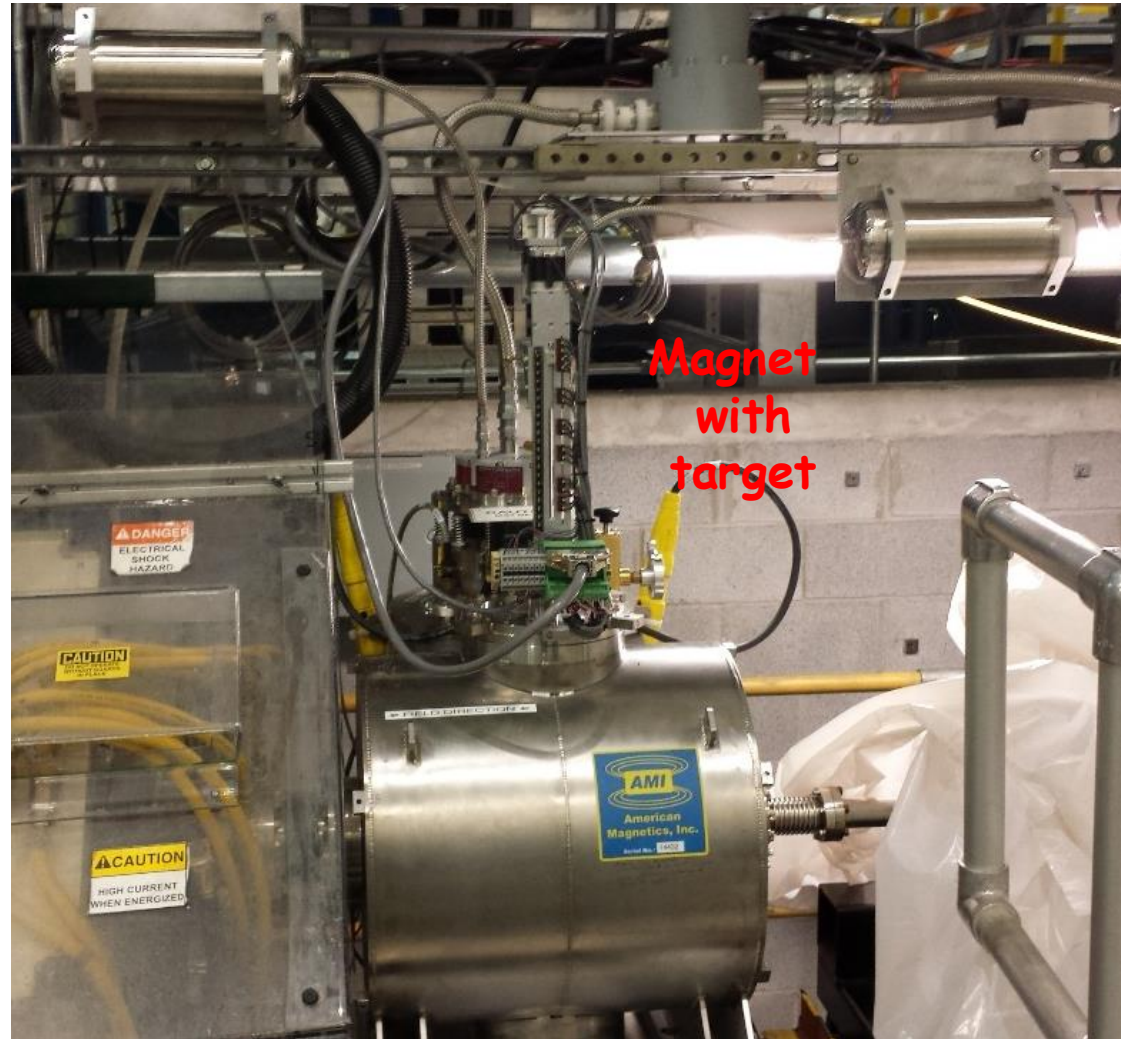


MØLLER SYSTEMATIC ERROR BUDGET

Variable	2010 (3.7T)	PREX (2017)
<i>Target polarization</i>	0.35%	0.25%
<i>Analyzing power</i>	0.3%	0.2%
<i>Levchuk effect</i>	0.3%	0.2%
<i>Target temperature</i>	0.02%	0.02%
<i>Dead time</i>	0.3%	0.05%
<i>Background</i>	0.3%	0.2%
<i>High beam current*</i>	0.2%*	0.2%
<i>Others</i>	0.5%	0.3%
<i>Total</i>	0.87%	0.53%

* M. Poelker et al. Phys.Rev.ST Accel.Beams 10 (2007) 053502

NEW MAGNET



New superconductive magnet:

Very limited information:

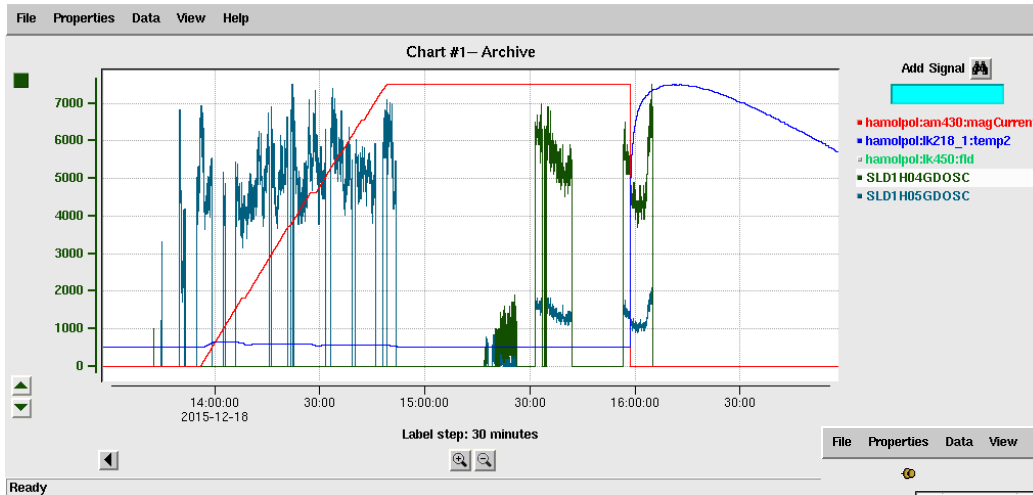
- Maximal field $\pm 5T$,
- Portable,
- No LN required,
- No cryo lines required,

Small amount of He →

low thermal capacity and “cold” production →

low field ramping gradient ($\sim 35\text{min}$: $0 \rightarrow 3T$)

FIRST EXPERIENCE WITH OPERATION



18 December, 2015

CW beam $1.5\mu\text{A}$

4T field in magnet

Target #3 ($12\mu\text{m}$)

Magnet **QUENCH**

5-6 hours to recovery

19 December, 2015

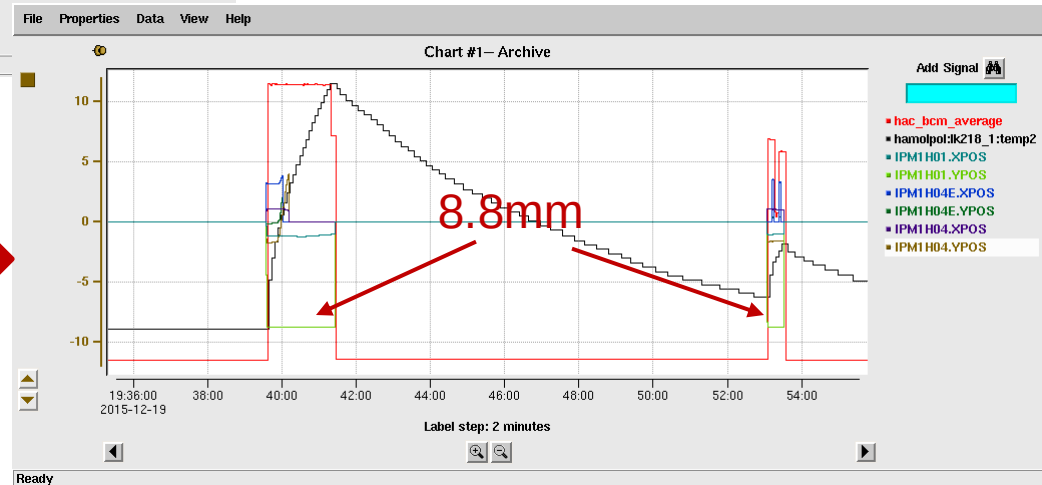
Beam through Compton

Pulse/Tune

Low radiation

No field in magnet

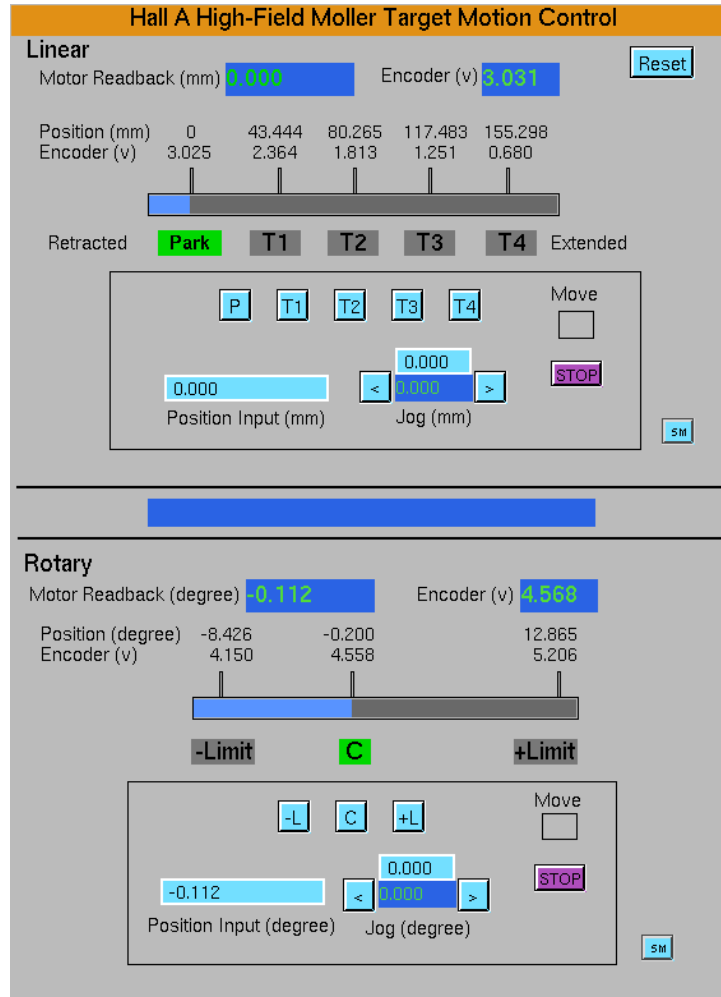
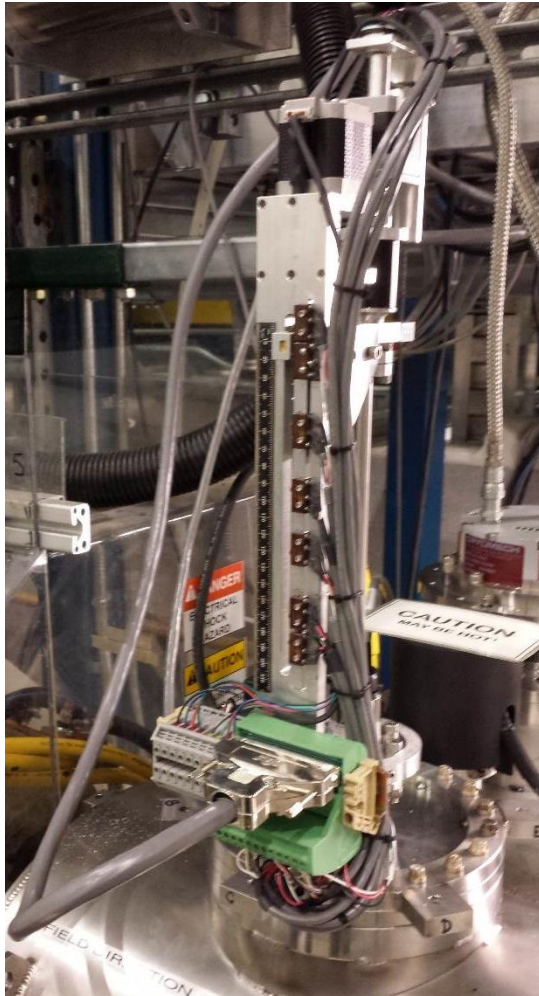
No Moller target



New SC magnet:

- ~12hours of CW, quench, lost of superconductivity,
- Need more experience,
- Magnet alignment check at low beam energy,
- Stronger beam tuning requirements,
- Do we need additional magnet shielding for PREX?

NEW POLARIZED ELECTRON TARGET



Designed and made by Temple University

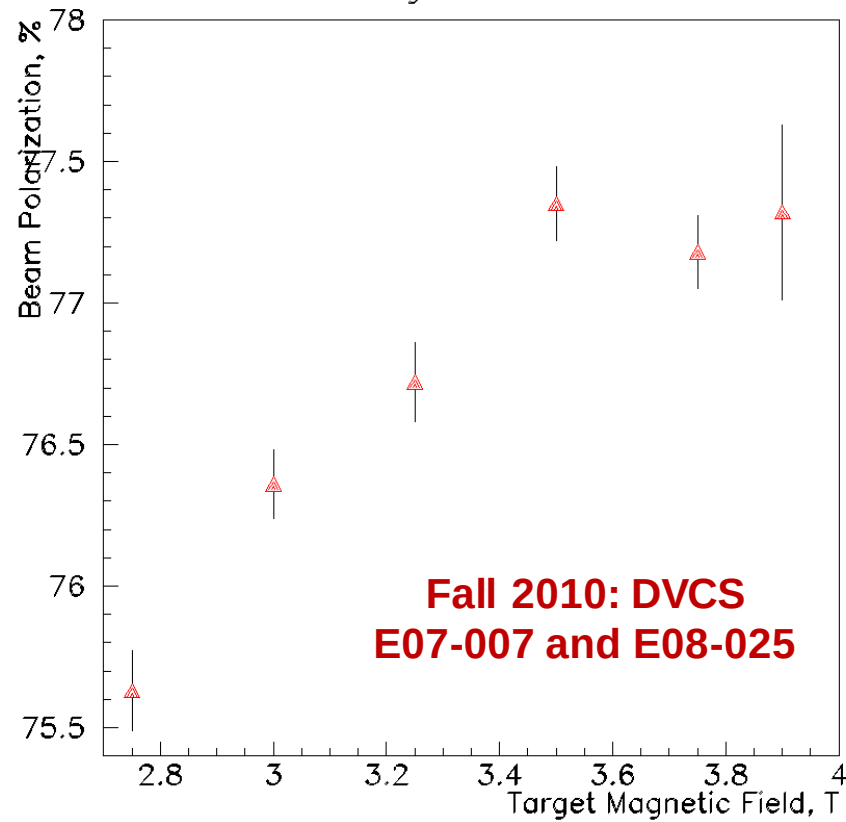
Rotation in horizontal plane $\pm 10^\circ$

Vertical translation

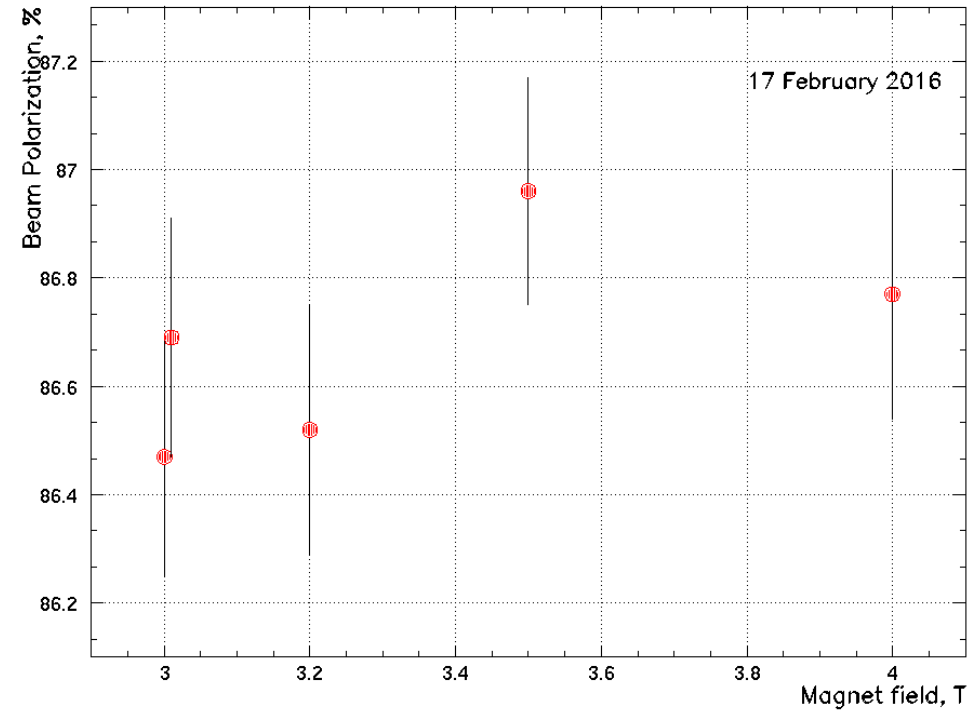
Targets in holder: pure iron $1\mu\text{m}$, $4\mu\text{m}$, $12\mu\text{m}$, $25\mu\text{m}$

NEW TARGET SATURATION TEST

DVCS Target Saturation Test



Measured Polarization vs. Target Saturation Field
Moller Polarimeter



MOLLER SPECTROMETER RELATED SYSTEMATICS

Analyzing power + Levchuk effect + magnets PS instability:

GEANT vs MEASUREMENT

Quads scan to study analyzing power and Levchuk effect

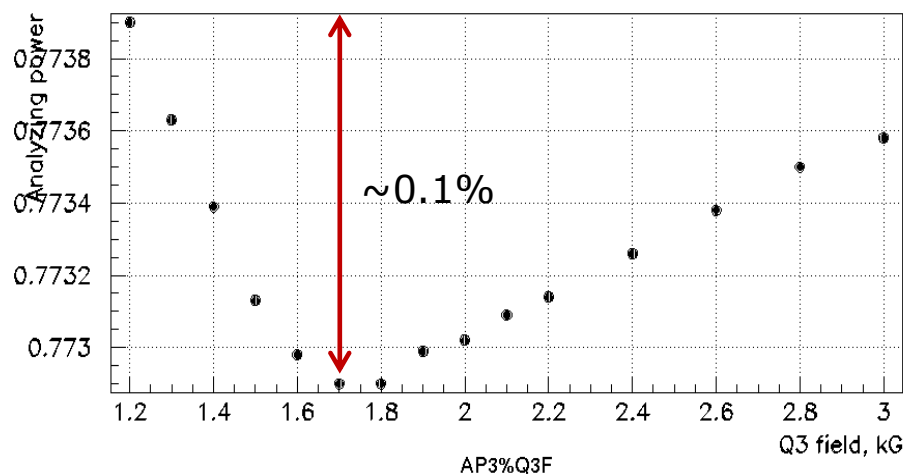
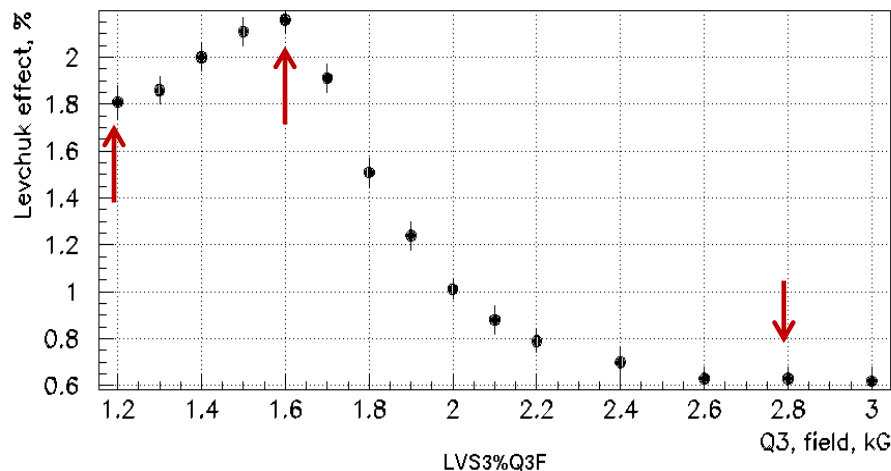
Dead time:

- measure it with using LED pulser

Combined optics:

Compton + Møller + Hall A target

- compatibility,
- centered beam on the Hall A target and on the Møller target,
- beam position lock for the Møller target?
- dithering vs. SC magnet turned on?



KERR EFFECT

The Kerr magneto-optic effect:

Was successfully used off-line in Hall C long time ago;

Unique facility to test the target mechanism/alignment with no beam

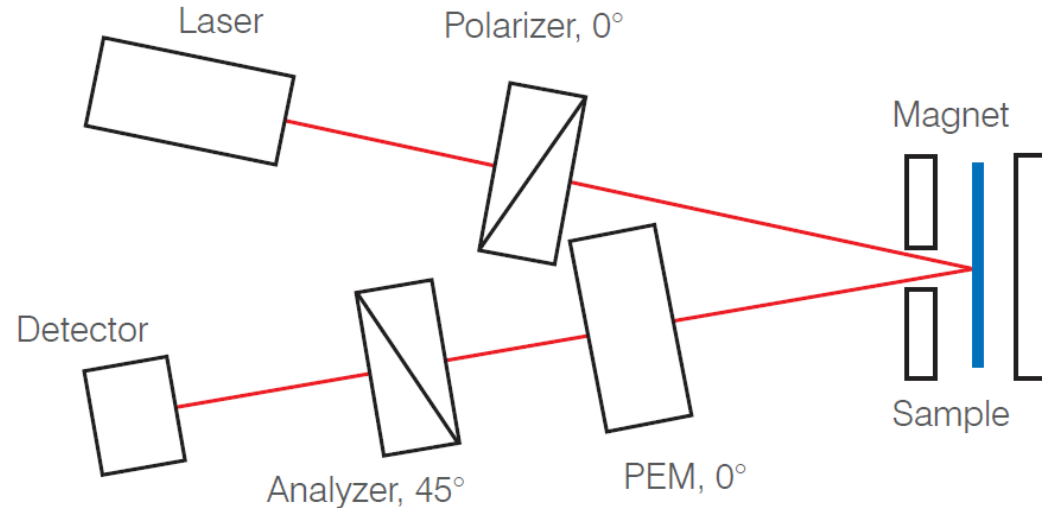
Polar Kerr effect is in one magnitude larger than Longitudinal or Transverse Kerr effect.

The magnetization is perpendicular to the surface, parallel to the direction Of the incident linearly polarized light: the polarization is rotated through an angle α_K given by

$$\alpha_K = K_R M$$

K_R - is the Polar Kerr constant

M – is the target magnetization



CONCLUSION

Magnet: *more experience with operation (spring - fall 2016)*

Target: *upgrade (spring - summer 2016)*

commissioning: off-line (Kerr) summer 2016?

on-line fall 2016

Spectrometer + DAQ:

GEANT simulation

GEANT vs measurements: spring - fall 2016

Per Aspera Ad Astra!

